From Scores to Steps:

Diagnosing and Improving LLM Performance in Evidence-Based Medical Calculations

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Group K23

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Based on Wang et al., 2026, arXiv v2.

Presentation Outline

1. Motivation and Research Problem
2. Evaluation Framework
3. MedRaC Methodology
4. Paper Experiments and Results
5. Local Demo and Mini-Experiment
6. Discussion and Conclusion

From a plausible score to a verifiable calculation process.

Medical Calculations Require More Than Arithmetic

Clinical notes

Relevant and irrelevant details coexist.

Creatinine clearance

Formula + patient variables

Clinical question

Select the calculator that fits the question.

Corrected sodium

Formula + unit-aware arithmetic

Clinical consequence

An incorrect score can change a decision.

Wells score

Clinical criteria + score

Medical calculation combines interpretation, selection, extraction, and computation.

Final-Answer Accuracy Can Hide Invalid Reasoning

Corrected sodium: reference

$$Na_{\text{corrected}} = Na_{\text{measured}} + 0.024(Glucose - 100) \quad (1)$$

Equation (1) gives 137.248 for $Na = 127$ and $Glucose = 527$.

Incorrect model formula

$$127 + 0.016(527) = 135.432 \quad (2)$$

Final-answer evaluation

1. Incorrect formula
2. Numerically plausible answer
3. Marked correct by broad tolerance

Formula: Incorrect

Final score accepted under the original tolerance.

A plausible answer is not necessarily a valid answer.

Research Questions and Contributions

Research questions

1. How can each reasoning stage be evaluated?
2. Where do LLM failures originate?
3. Can external tools reduce these errors?

Contributions

1. Step-wise evaluation
2. Structured error attribution
3. MedRaC: Formula RAG + Python execution

The paper evaluates, diagnoses, and then intervenes.

Medical Calculation as a Four-Stage Process

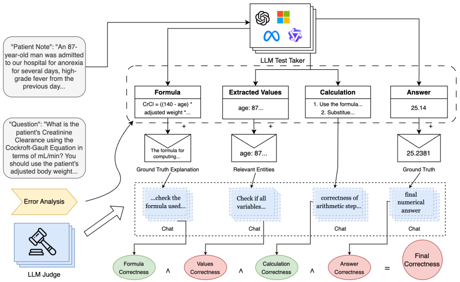

Figure 1: Four-stage evaluation of medical calculations.

F: Formula	E: Extraction	C: Calculation	A: Answer
calculator method	patient values	valid operations	reference result

Each stage can fail independently and propagate errors downstream.

Strict Correctness Requires Every Stage to Pass

Case-level criterion

$$\kappa = V(F) \wedge V(E) \wedge V(C) \wedge V(A) \tag{3}$$

$V(\cdot)$
Binary validation of a stage.

κ
Overall correctness for one case.

A correct case requires a valid Formula, Extraction, Calculation, and Answer.

Conditional Correctness Isolates Stage Reliability

Can stage S_i succeed after valid prerequisites?

$$CC_i = P(V(S_i) \mid V(S_1) \land \cdots \land V(S_{i-1})) \tag{4}$$

S_i
The current stage in the F–E–C–A sequence.

Condition
All earlier stages must already be valid.

Conditional Correctness separates a stage's reliability from upstream failures.

First Error Attribution Identifies the Earliest Failure

Where does an incorrect case first break?

$$FE_i = P(V(S_1) \wedge \cdots \wedge V(S_{i-1}) \wedge \neg V(S_i) \mid \neg \kappa) \quad (5)$$

$\neg V(S_i)$
Stage S_i is invalid.

$\neg \kappa$
The overall case is incorrect.

First Error Attribution directs improvement to the first failed capability.

Error Taxonomy Turns Failures into Diagnoses

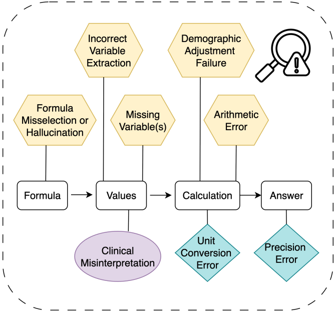

Figure 2: Error categories in medical calculations.

Knowledge
Formula selection; hallucination; demographic rules

Clinical interpretation
Incorrect extraction; missing variables; misinterpretation

Numerical execution
Unit conversion; arithmetic; rounding

Error attribution identifies which capability requires improvement.

MedRaC Grounds Formulas and Executes Calculations

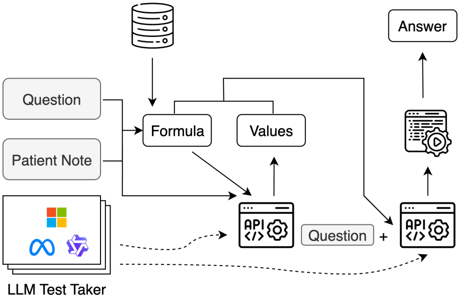

Figure 3: MedRaC architecture.

Retrieve formula	Extract values	Generate code	Execute code
grounding	clinical text	formula to program	deterministic result

The LLM interprets clinical text; tools ground formulas and compute deterministically.

Each MedRaC Component Targets a Distinct Failure Mode

Table 1: MedRaC components and targeted failures.

Component	Targeted failure
Formula RAG	Formula misselection and hallucination
Value extraction	Missing or incorrect inputs
Code generation	Formula-to-computation translation
Python execution	Arithmetic errors

Interpretation-dependent errors remain difficult.

Experimental Setup for Calculation and Clinical Scoring

Table 2: Paper experimental setup.

Benchmark	55 medical calculators; 1,048 original cases, 940 after cleaning
Task types	Equation-based and rule-based
Methods	Direct, CoT, One-shot, Self-Refine, MedPrompt, MedRaC
Models	Multiple open and closed LLMs
Evaluation	Final-answer and LLM step-wise evaluation

The benchmark tests both deterministic calculation and clinical scoring.

Main Results Across Models and Prompting Methods

Model	Direct		CoT		One-shot		Self-Refine		Medprompt		MedRaC	
	Rule	Calc	Rule	Calc	Rule	Calc	Rule	Calc	Rule	Calc	Rule	Calc
Phi-4-mini	16.52	2.16	5.01	2.16	12.09	16.47	2.06	6.16	3.24	10.32	**35.10**	**68.39**
LLaMA3.2-3B	12.39	0.83	1.47	3.99	**14.75**	**18.47**	0.88	2.83	2.95	16.31	-[2]	-
Qwen3-4B	23.30	26.79	9.73	25.79	**49.85**	59.57	9.44	26.29	10.32	45.92	45.72	**68.72**
Qwen3-8B	29.20	42.26	16.52	38.10	**58.70**	62.90	19.76	40.10	15.93	53.74	46.61	**74.54**
LLaMA3.1-8B	19.17	2.50	6.78	8.32	25.07	20.97	5.90	6.82	11.21	28.45	**31.27**	**70.22**
Qwen3-14B	39.53	46.59	26.55	43.43	**60.77**	67.05	27.14	47.25	8.55	22.30	50.44	**78.37**
GPT-4o-mini	22.42	7.99	23.89	34.61	**52.80**	49.58	26.55	33.94	11.80	42.43	50.44	**72.71**
GPT-4o	24.48	13.98	43.07	43.93	**62.24**	54.24	44.84	44.93	25.96	56.91	51.03	**64.39**

Table 3: Main performance results across models and methods.

Overall main result

MedRaC consistently improves calculation accuracy across the reported models, with the largest gains for smaller open models. Rule-based results are mixed, confirming that clinical interpretation remains the limiting capability.

Main Result: MedRaC Improves Equation-Based Tasks

Table 4: Paper-reported equation-based accuracy (%).

Model	One-shot	MedRaC	Gain
Phi-4-mini	16.47	**68.39**	**+51.92**
LLaMA3.1-8B	20.97	**70.22**	**+49.25**
GPT-4o	54.24	**64.39**	**+10.15**

Main finding

Formula grounding and deterministic execution increase equation-based accuracy by **10.15–51.92 percentage points**.

Rule-based gains remain mixed because clinical interpretation still dominates scoring.

Step-Wise Evaluation Aligns Closely with Expert Judgment

Table 5: LLM–Expert agreement across evaluation stages (%).

Agreement type	Formula	Extraction	Calculation	Answer
Expert–Expert	84.8	84.8	89.1	95.7
Expert–Non-Expert	72.3	78.1	66.6	91.3
LLM–Expert	**90.2**	78.3	88.1	**97.8**

Validation set

46 samples; five calculators.

Interpretation

Alignment is strongest for Formula and Answer; Extraction remains the hardest stage.

Error Reductions Reveal MedRaC's Remaining Boundary

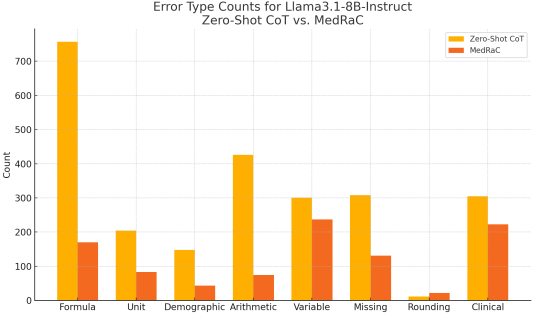

Figure 4: Error reduction with MedRaC.

Large reductions

Formula

Arithmetic

Demographic adjustment

Remaining boundary

Incorrect extraction

Clinical misinterpretation

MedRaC solves tool-addressable errors better than interpretation-dependent errors.

Local Live Demo Makes the F–E–C–A Pipeline Observable

Figure 5: Local live-demo pipeline.

Inspect
Intermediate outputs

Demonstrate
Complete pipeline

Compare
Plain CoT vs. MedRaC

The live run connects paper stages to observable outputs.

Local Mini-Experiment

Table 6: Fixed 10-sample comparison.

Metric	Plain CoT	MedRaC + RAG
Final-answer accuracy	9/10	**10/10**
Formula correctness	10/10	10/10
Extraction correctness	9/10	**10/10**
Calculation correctness	10/10	10/10
Strict all-steps correctness	8/10	**9/10**

Same fixed samples, baseline, and proposed pipeline

MedRaC improved extraction and strict all-steps correctness in this local run.

Discussion

Limitations

- Structured, English-only cases
- Dependence on formula-bank quality
- Dependence on an LLM Judge
- Limited human validation

Future work

- Stronger clinical interpretation
- Broader human validation
- Richer formula knowledge
- Human-in-the-loop evaluation

Reliable medical AI requires tools and clinically grounded interpretation.

Conclusions

1. **Final-answer accuracy can mask clinically relevant reasoning errors.**
2. **Step-wise evaluation makes Formula, Extraction, Calculation, and Answer failures measurable.**
3. **MedRaC improves formula grounding and arithmetic reliability; clinical interpretation remains open.**

Reliable medical AI requires correct outputs and verifiable processes.

Thank You — Questions and Discussion

Thank you

Questions and Discussion

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